Seth Ricks ECEN 380: PS11 3/24/2026

7.17(d)(f), 7.18(d), 7.24(a)(d), MATLAB 7.54, 7.55

7.17 (d) Determine the z-transform and ROC for the following time signal:

$\boxed{1/5}$

(d) $x[n] = (\frac{1}{4})^n (u[n] - u[n-5])$

$x[n] = (\frac{1}{4})^n u[n] - (\frac{1}{4})^n u[n-5] = (\frac{1}{4})^n u[n] - \left[(\frac{1}{4})^{n-5} u[n-5]\right](\frac{1}{4})^5$

E.1: $\mathcal{Z}\{\alpha^n u[n]\} = \frac{1}{1-\alpha z^{-1}}$, ROC: $|z| > |\alpha|$

define: $a[n] = (\frac{1}{4})^n u[n] \longrightarrow A[z] = \frac{1}{1-\frac{1}{4}z^{-1}}$, ROC: $|z| > |\frac{1}{4}|$

E.2: $\mathcal{Z}\{x[n-k]\} = z^{-k}X(z)$, ROC: $R_x$, except possibly $|z| = 0, \infty$

define: $b[n] = (\frac{1}{4})^{n-5} u[n-5] = a[n-5]$

$\hookrightarrow B[z] = \frac{z^{-5}}{1-\frac{1}{4}z^{-1}}$

define: $c[n] = (\frac{1}{4})^5 b[n] \longrightarrow C[z] = \frac{(\frac{1}{4})^5 z^{-5}}{1-\frac{1}{4}z^{-1}}$

$X[z] = A[z] + C[z]$

$X[z] = \frac{1}{1-\frac{1}{4}z^{-1}} - \frac{(\frac{1}{4})^5 z^{-5}}{1-\frac{1}{4}z^{-2}} = \boxed{X[z] = \frac{1-(\frac{1}{4})^5 z^{-5}}{1-\frac{1}{4}z^{-1}}, \text{ ROC: } |z| > |\frac{1}{4}|}$

7.17(f) Determine z-transform and ROC for the following time signal: $\boxed{2/5}$

$$x[n] = 3^n u[-n-1]$$

E.1: $\mathcal{Z}[\alpha^n u[n]] = \frac{1}{1-\alpha z^{-1}}$, ROC: $|z| > |\alpha|$

E.2: $\mathcal{Z}\{x[-n]\} = X(1/z)$, ROC: $1/R_x$

E.3: $\mathcal{Z}\{x[n-k]\} = z^{-k}X(z)$, ROC: $R_x$, except possibly $|z| = 0, \infty$

$x[n] = 3^n u[-n-1] = -(-3^n u[-n-1])$

E.1.1: $\mathcal{Z}\{-\alpha^n u[-n-1]\} = \frac{1}{1-\alpha z^{-1}}$, ROC: $|z| < |\alpha|$

$\boxed{X[z] = -\frac{1}{1-3z^{-1}}, \text{ ROC: } |z| < 3}$

7.18(d) Given each of the following z-transforms, determine whether the DTFT of the corresponding time signal exists without determining the signal, and identify the DTFT in those cases where it does exist. 3/5

(d) $X(z) = \dfrac{z^{-1}}{(1-\frac{1}{2}z^{-1})(1+3z^{-1})}, \quad \frac{1}{2} < |z| < 3$

page 556: "Existence of DTFT requires absolute summability of $x[n]$"

slides: absolute summability:

$$\sum_{n=-\infty}^{\infty} |x[n]| r^{-n} < \infty$$

notecard

DTFT exists, because ROC includes 1 (the unit circle)

Page 557: DTFT $= X(z)\big|_{z=e^{rj\Omega}}$, where $r = 1$

$$X(e^{j\Omega}) = \frac{(e^{j\Omega})^{-1}}{(1-\frac{1}{2}(e^{j\Omega})^{-1})(1+3(e^{j\Omega})^{-1})}$$

$$\boxed{X(e^{j\Omega}) = \frac{e^{-j\Omega}}{(1-\frac{1}{2}e^{-j\Omega})(1+3e^{-j\Omega})}}$$

7.24 (a)

Use the method of partial fractions to obtain the time-domain signal to the z-transform:

4/5

(a) $$X(z) = \frac{1 + \frac{7}{6} z^{-1}}{(1 - \frac{1}{2} z^{-1})(1 + \frac{1}{3} z^{-1})}, \quad |z| > \frac{1}{2}$$

$$\frac{1 + \frac{7}{6} x}{(1 - \frac{1}{2} x)(1 + \frac{1}{3} x)} = \frac{A}{1 - \frac{1}{2} x} + \frac{B}{1 + \frac{1}{3} x}$$

$$1 + \frac{7}{6} x = A(1 + \tfrac{1}{3} x) + B(1 - \tfrac{1}{2} x)$$

$$1 + \frac{7}{6} x = A + \frac{A}{3} x + B - \frac{B}{2} x$$

$$\begin{aligned} A + B &= 1 \\ \tfrac{1}{3} A - \tfrac{1}{2} B &= \tfrac{7}{6} \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 1 \\ \frac{1}{3} & -\frac{1}{2} & \frac{7}{6} \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$$

$A = 2$

$B = -1$

$$= \frac{2}{1 - \frac{1}{2} z^{-1}} - \frac{1}{1 + \frac{1}{3} z^{-1}}$$

$$= 2\left(\frac{1}{1 - \frac{1}{2} z^{-1}}\right) + -1\left(\frac{1}{1 - -\frac{1}{3} z^{-1}}\right)$$

$$\boxed{x[n] = 2(\tfrac{1}{2})^n u[n] - (-\tfrac{1}{3})^n u[n]}$$

7.24 (d) Use method of partial fractions to find time-domain signal for this z-trans: $\boxed{5/5}$

(d) $X(z) = \dfrac{z^2 - 3z}{z^2 + \frac{3}{2}z - 1}, \quad \frac{1}{2} < |z| < 2$

$$X(z) = \frac{z^{-2}}{z^{-2}} \cdot \frac{z^2 - 3z}{z^2 + \frac{3}{2}z - 1} = \frac{1 - 3z^{-1}}{1 + \frac{3}{2}z^{-1} - z^{-2}}$$

partials: $\dfrac{-(1-3x)}{x^2 - \frac{3}{2}x - 1} = \dfrac{A}{(x + \frac{1}{2})} + \dfrac{B}{(x-2)}$

$$-1 + 3x = Ax - 2A + Bx + \tfrac{1}{2}B$$

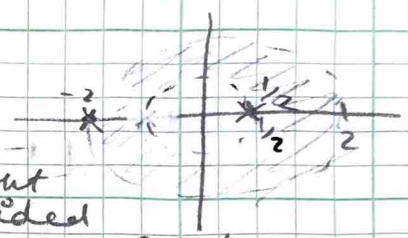
$A + B = 3$

$-2A + \frac{1}{2}B = -1$ $\quad \begin{bmatrix} 1 & 1 & 3 \\ -2 & \frac{1}{2} & -1 \end{bmatrix} \xrightarrow{RREF} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

$$= \frac{1}{x + \frac{1}{2}} + \frac{2}{x-2} = \frac{1}{\frac{1}{2} - z^{-1}} + \frac{2}{-(2 - z^{-1})} = \frac{1}{\frac{1}{2}(1 - 2z^{-1})} + \frac{2}{-2(1 - \frac{1}{2}z^{-1})}$$

$$= \frac{2}{1 - -2z^{-1}} + \frac{-1}{1 - \frac{1}{2}z^{-1}}$$

Interest: pole: $z = -2$ pole: $z = \frac{1}{2}$




* ROC outside outermost pole → right sided

ROC inside innermost pole → left sided

ROC between poles → two sided (one term right, one left)

first term – ROC: $|z| < 2$

second term – ROC: $|z| > \frac{1}{2}$

$$\boxed{x[n] = \underbrace{-(-2)^n u[-n-1]}_{\text{table E.1.1}} - \underbrace{(\tfrac{1}{2})^n u[n]}_{\text{table E.2}}}$$

# MATLAB Problem 7.54

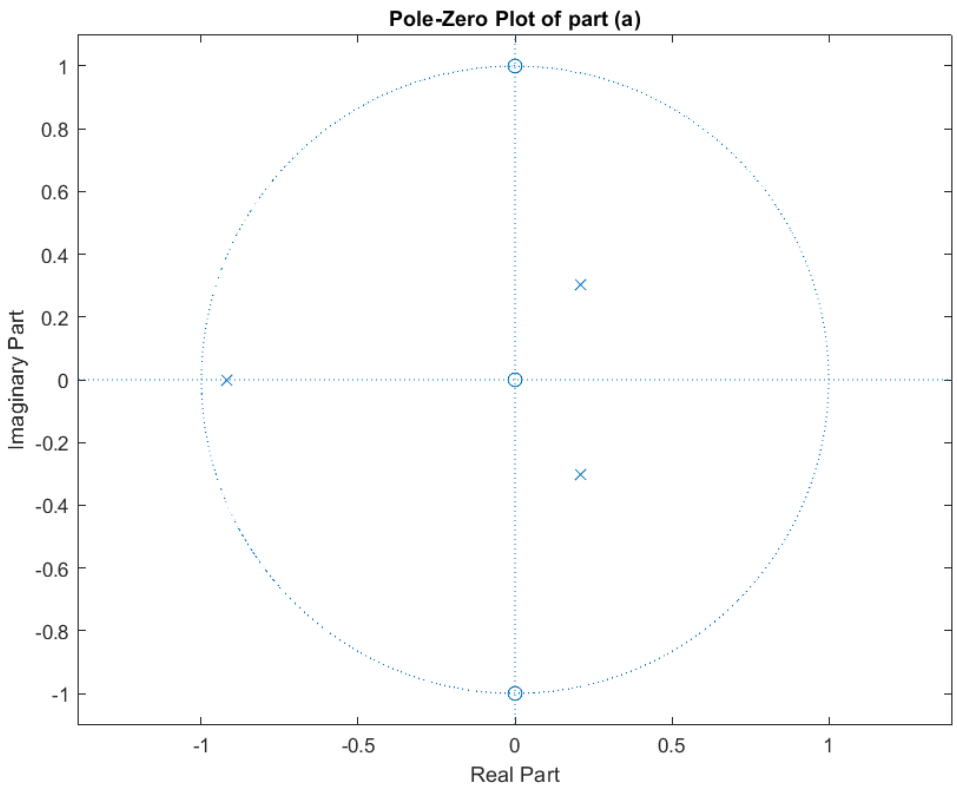
Use the MATLAB command **zplane** to obtain a pole-zero plot for the following systems:

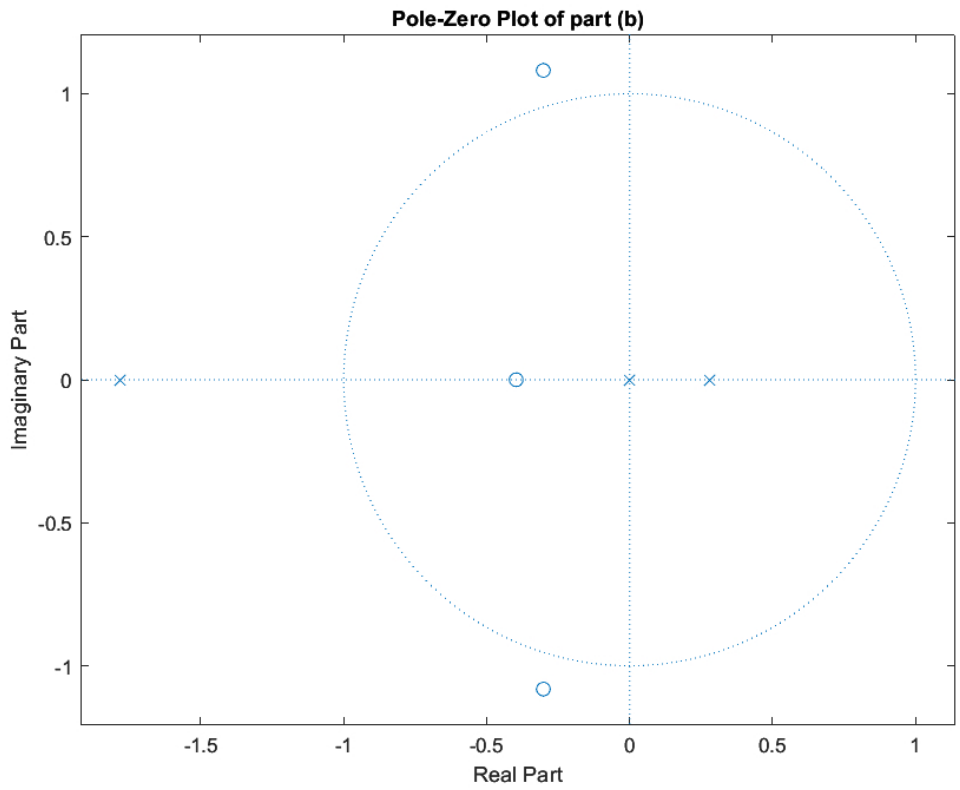
(a) $H(z) = \dfrac{1 + z^{-2}}{2 + z^{-1} - 1/2z^{-2} + 1/4z^{-3}} = \dfrac{z^3 + z}{2z^3 + z^2 - 1/2z + 1/4}$

(b) $H(z) = \dfrac{1 + z^{-1} + 3/2z^{-2} + 1/2z^{-3}}{1 + 3/2z^{-1} - 1/2z^{-2}} = \dfrac{z^3 + z^2 + 3/2z + 1/2}{z^3 + 3/2z^2 - 1/2z}$

```
% (a)
figure(2); clf; clear; clc; format compact
num = [1 0 1 0];
den = [2 1 -1/2 1/4];
zplane(num,den)
title('Pole-Zero Plot of part (a)')

% (b)
figure(2); clf; clear; clc; format compact
num = [1 1 3/2 1/2];
den = [1 3/2 -1/2 0];
zplane(num,den)
title('Pole-Zero Plot of part (b)')
```


Pole-Zero Plot of part (a)
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1


Pole-Zero Plot of part (b)
Imaginary Part
Real Part
1
0.5
0
-0.5
-1
-1.5
-1
-0.5
0
0.5
1

# MATLAB Problem 7.55

Use the MATLAB command **residuez** to obtain the partial-fraction expansions required to solve Problem 7.24(d)-(g).

(d) $X(z) = \dfrac{z^2 - 3z}{z^2 + 3/2z - 1}, 1/2 < |z| < 2$

(e) $X(z) = \dfrac{3z^2 - 1/4z}{z^2 - 16}, |z| > 4$

(f) $X(z) = \dfrac{z^3 + z^2 + 3/2z + 1/2}{z^3 + 3/2z^2 + 1/2z}, |z| < 1/2$

(g) $X(z) = \dfrac{2z^4 - 2z^3 - 2z^2}{z^2 - 1}, |z| > 1$

```
format compact
clc
clear
num=[1 -3]
den=[1 3/2 -1]
[R, P, K] = residuez(num,den)

format compact
clc
clear
num=[3 -1/4]
den=[1 0 -16]
[R, P, K] = residuez(num,den)
```

```
format compact
clc
clear
num=[1 1 3/2 1/2]
den=[1 3/2 1/2]
[R, P, K] = residuez(num,den)

format compact
clc
clear
num=[2 -2 -2]
den=[1 0 -1]
[R, P, K] = residuez(num,den)
```

```
num = 1x2
       1.0000e+000    -3.0000e+000

den = 1x3
       1.0000e+000     1.5000e+000    -1.0000e+000

d_R = 2x1
       2.0000e+000
      -1.0000e+000

d_P = 2x1
      -2.0000e+000
     500.0000e-003

d_K =
     []
```

```
d_K =
     []
num = 1x2
       3.0000e+000  -250.0000e-003

den = 1x3
       1.0000e+000     0.0000e+000   -16.0000e+000

e_R = 2x1
       1.4688e+000
       1.5312e+000

e_P = 2x1
       4.0000e+000
      -4.0000e+000
```

```
num = 1x4
       1.0000e+000     1.0000e+000     1.5000e+000   500.0000e-003

den = 1x3
       1.0000e+000     1.5000e+000   500.0000e-003

f_R = 2x1
       2.0000e+000
      -1.0000e+000

f_P = 2x1
      -1.0000e+000
    -500.0000e-003

f_K = 1x2
       0.0000e+000     1.0000e+000
```

```
num = 1x3
       2.0000e+000    -2.0000e+000    -2.0000e+000

den = 1x3
       1.0000e+000     0.0000e+000    -1.0000e+000

g_R = 2x1
       1.0000e+000
      -1.0000e+000

g_P = 2x1
      -1.0000e+000
       1.0000e+000

g_K =      2.0000e+000
```